Forecasting and Macroeconomic Analytics

ECON 522

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Cycles

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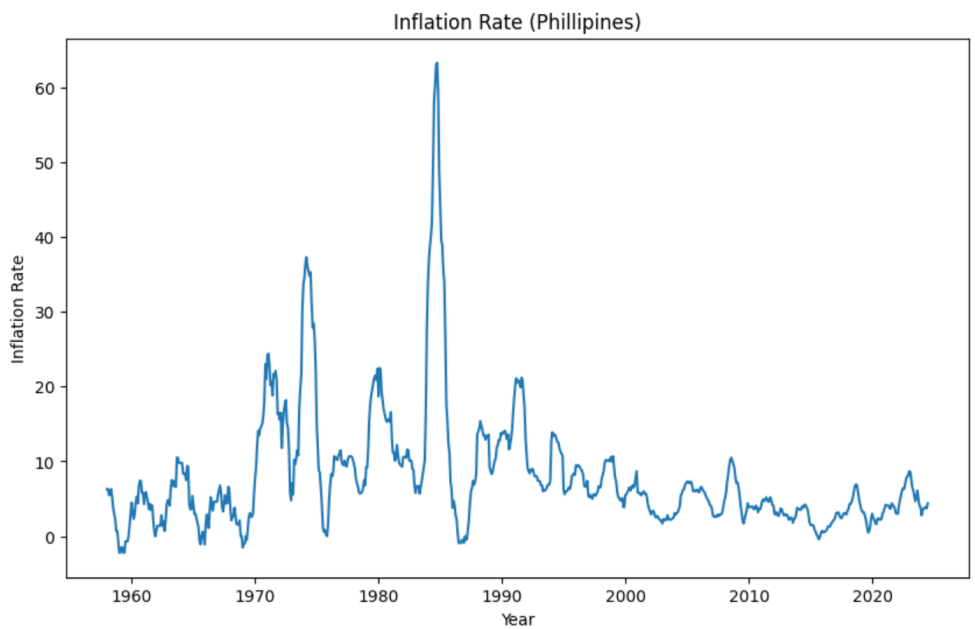
- Cycles include any sort of dynamics that is not captured by trend and seasonality.
- This includes dynamic, persistence, and any way in which the present is linked to the past or the future.
- We will spend most of the rest of this class to discuss which models to use to estimate and forecasting the cycles component.

Cycles

- We can explain this as thinking that the error terms/data are positively correlated with each other over time.
- One good year will tend to be followed by another, and one bad year will be followed by another bad year.
- Eventually, of course, something unusual will happen and a bad year will be followed by a good year. This represents the end of the recession and the start of the next boom.
- The economy is subject to *shocks*! These shocks move the economy up or down in the period that occurs, then persist for a period of time.
- The shocks are represented by the values of the error terms.
- Think of technology shocks slowly moving to different sectors.

Autocorrelation

When we think of cycles, we often think about *autocorrelation*, the correlation of a series with itself lagged.

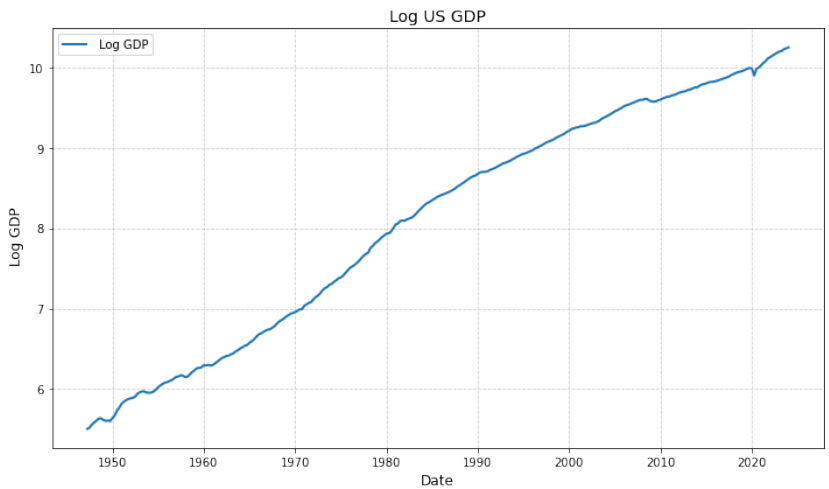


very high

	Inflation Rate	Change in the Inflation Rate
Lag 1	0.983451	0.443811
Lag 2	0.952210	0.311938
Lag 3	0.910634	0.233348
Lag 4	0.861315	0.186828

- The inflation rate is **highly serially correlated** since $\rho(1) = 0.98$
- Last month's inflation rate contains much information about this month's inflation. Moreover, the plot is dominated by multiyear swings but there are still surprise movements!

Autocorrelation of log GDP

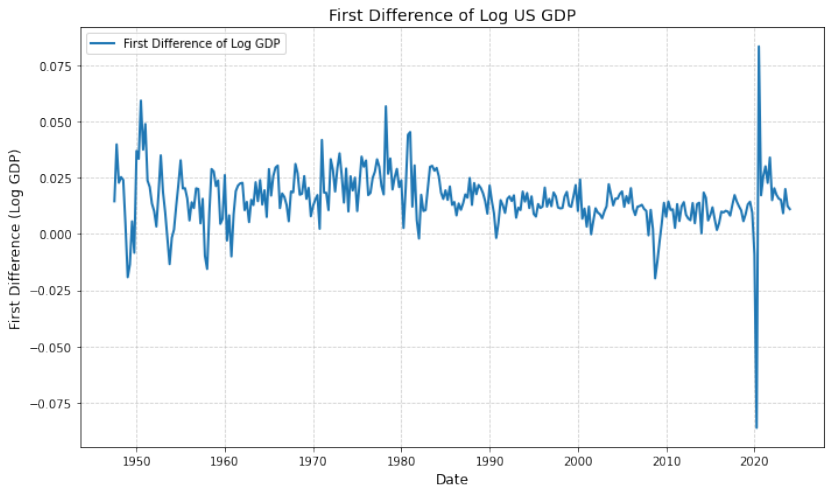


Lag	Autocorrelation
0	1.000000
1	0.991150
2	0.982262
3	0.973444
4	0.964663
5	0.955902
6	0.947178
7	0.938413
8	0.929516
9	0.920517
10	0.911469

- log GDP is VERY persistent.
- A trend is a persistent component of a series and log GDP has a trend.
- I could have gone longer. At lag 20 it was still 0.82.

Autocorrelation of GDP growth

Recall the GDP growth data



Lag	Autocorrelation
0	1.000000
1	0.261227
2	0.254689
3	0.098059
4	0.030842
5	-0.015595

- GDP growth is much less correlated than GDP. Taking the log difference has eliminated some of the persistence.
- However, that 0.26 and 0.25 is still significantly different than zero (we will see more on this later).

How do we model cycles?

- The best way to capture the fact that things move slowly is to include a lagged dependent variable as an explanatory variable (ala $AR(1)$). *Why the name?*
- This is what we did when we were modeling the model of GDP with trend etc.
- I could also add two lags $\implies AR(2)$
- Or we can use *MA* or *ARMA* models! We will learn to see which model/s are best in the next few classes.

Cycles - in Practice

- Most time series will need **at least** an $AR(1)$, often not more than an $AR(2)$
- Once you add lags and other variables, the coefficients may become insignificant. Intuition?
- R^2 will increase significantly!
- AR terms are your best friends when it comes to forecasting.
- They are at the core of time series analysis.

Stationarity

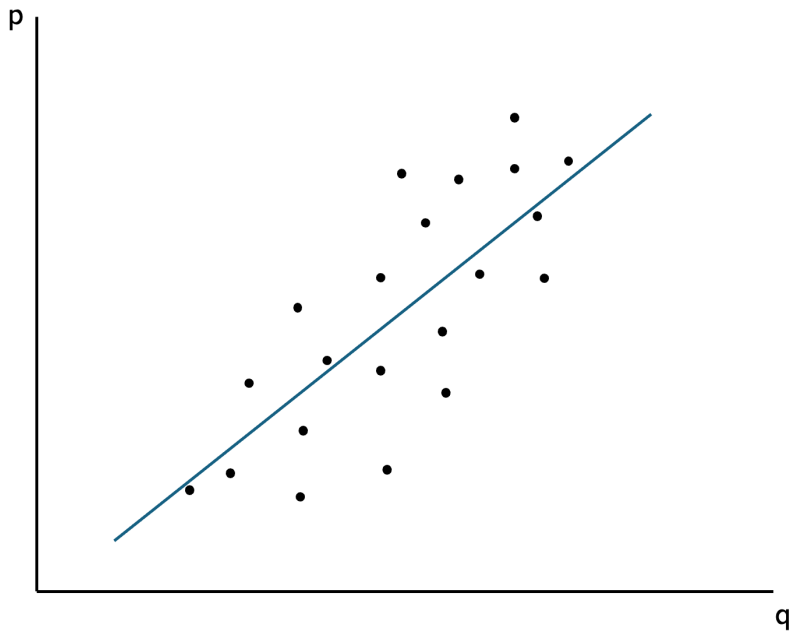
- Before modeling cycles, we need to define when a time series is *stationary* since we can only deal (for the most part) with data that is stationary.
- Think about a time series. We observe a part of the path of the series. In theory, a time series begins in the infinite past and continues in the infinite future and we only observe a small part of it.
- Since we only observe a finite period, we would like the series to be stable over all periods.
- **Stationarity** is then very important! We will discuss the exact definition later.

Stationarity Intuition

- Shortly we'll be building models for cyclical time series,i.e. we will learn to move the covariance/correlation structure of the data.
- If the underlying probabilistic structure of the series were changing over time, we'd be doomed - there would be no way to relate the future to the past, because the laws governing the future would differ from those governing the past.
- In this sense, we cannot forecast a structural break. We ca model it and forecast the detrended data.
- We will talk a lot more about all this. First let's talk more about which questions we can ask with time series analysis

Structural vs Time Series Models

Simple Intuition



- If we regress p on q, the estimated regression line will be the blue line. Is this the *demand curve* or the *supply curve*?

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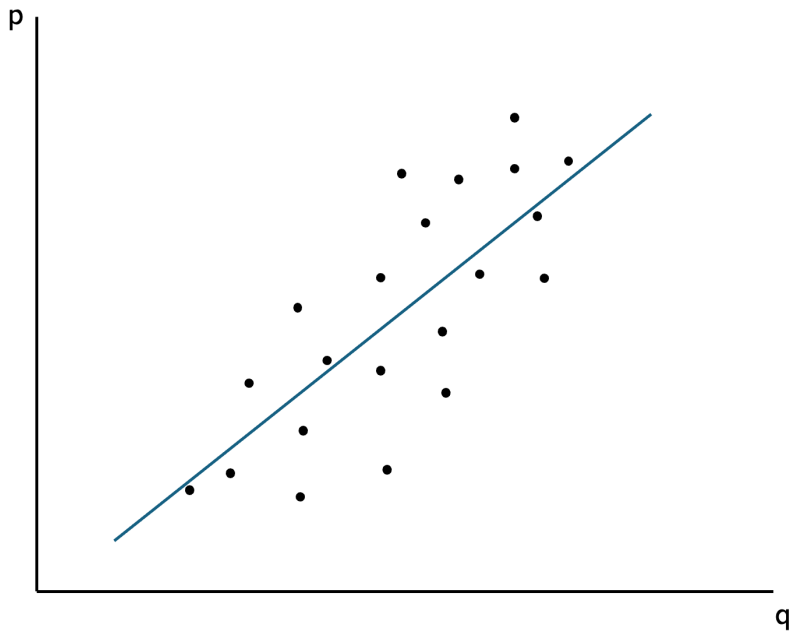

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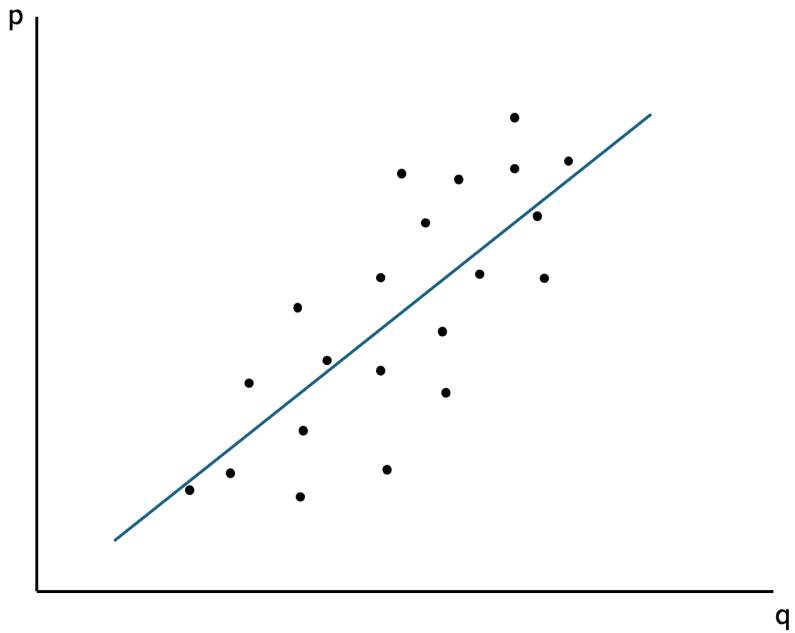

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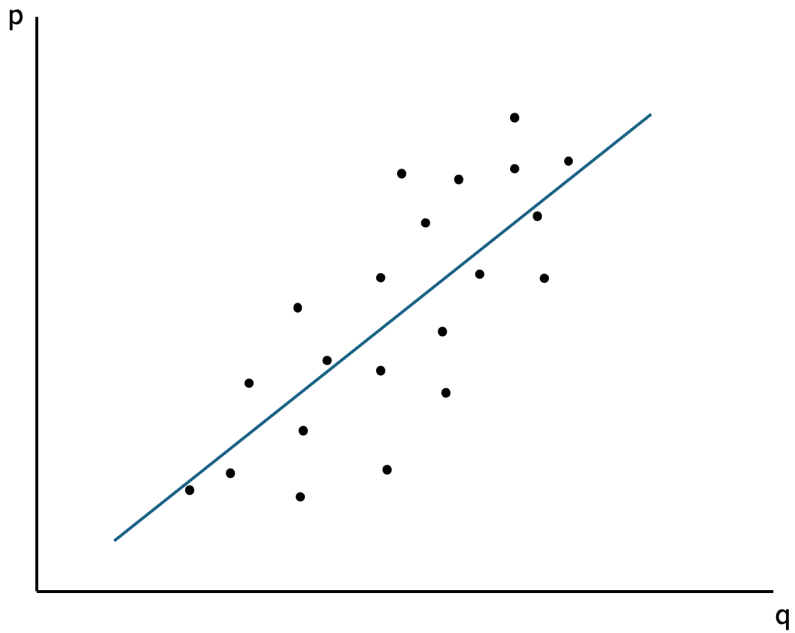
- The estimated slope is not the slope of the demand function but a combination of the true slope of demand and supply
- The estimator is not correct, it is biased! The slope is not estimating the elasticity. There is a simultaneity bias!

Simple Intuition

OLS assumes regressor is exogenous
– for forecasting – it doesn't matter



does it matter?
what is your question

- If we regress p on q, the estimated regression line will be the blue line. Is this the *demand curve* or the *supply curve*? neither – combination
- The estimated slope is not the slope of the demand function but a combination of the true slope of demand and supply
- The estimator is not correct, it is biased! The slope is not estimating the elasticity. There is a simultaneity bias!
- The prices are a **combined** effect of supply and demand, because both are price contingent but also influence prices *simultaneously!*

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  - You will not be estimating the elasticity by simply regressing demand on prices
- If we really want to estimate the slope of the demand function, we need to make sure that the RHS variable is "exogenous/predetermined" (e.g. say weather)

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- What you do is a combination of these two which introduces difficulties in interpreting your coefficients.

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Think abt what question you're asking

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- If you are not sure, then DO NOT give too much weight on the estimates of price elasticity, as a bias is probably there!
- We cannot always give a **causal** interpretation unless we are sure that p is exogenous
- This will not affect the quality of your forecasts!